

6th Oct

Case 1 $\rightarrow$ due tonight

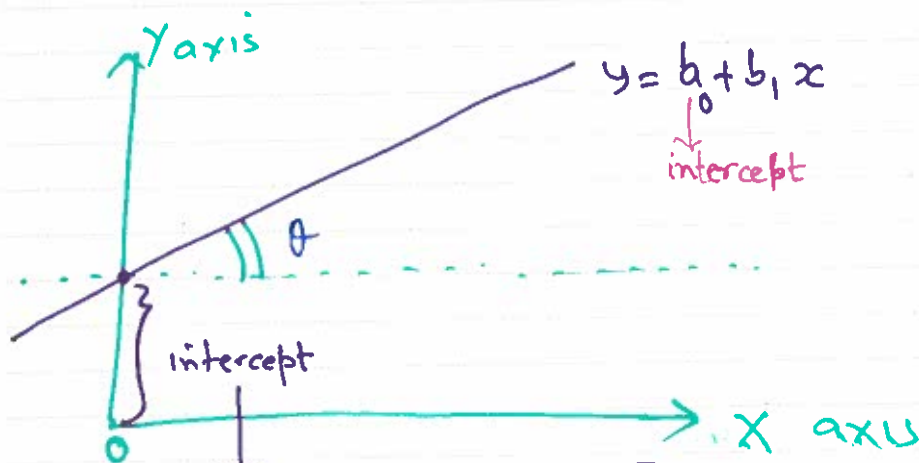
Lecture: Regression

Diamond Example

Similar clarity: VS1, VS2, etc

✓ Response: Price (\$)

X Explanatory variable: Weight (Carats)



$$y|_{x=0} = b_0 + b_1(0) = \boxed{b_0}$$

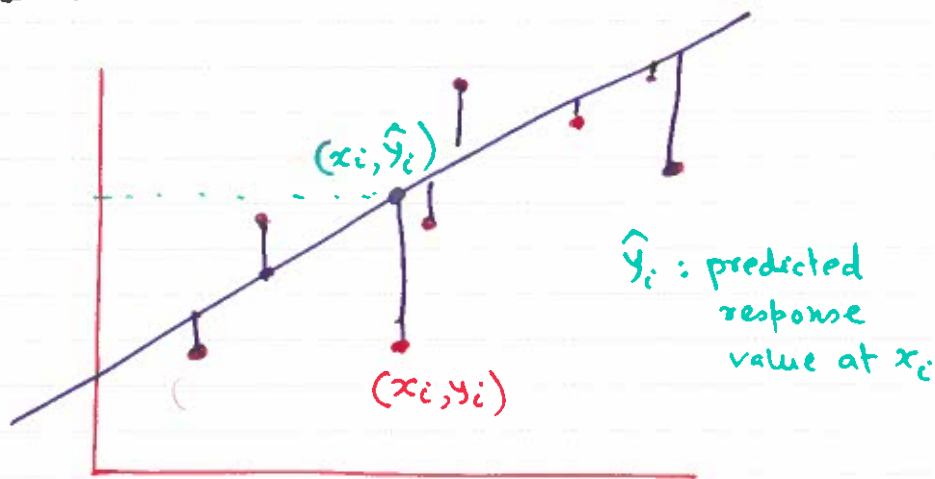
$b_1 = \text{slope of the line} = \tan(\theta)$
 $\quad \quad \quad := \tan(\theta)$

i	X	Y
1	x_1	y_1
2	x_2	y_2
.	.	.
1	.	.
i	x_i	y_i
.	.	.
n	.	.

$$e_i = y_i - \hat{y}_i$$

$$\text{Sum } e_i^2$$

Goal: Fit ^{the} best possible line through data



$$\text{residual} = \hat{e}_i = y_i - \hat{y}_i$$

i: observation number (excel: row number)

$\hat{e}_i$ are deviations from fitted line

"best" $\rightarrow$ Least Squares

selects b_0, b_1 such that sum of squared

intercept slope

deviations of the response (y) from the line is minimised

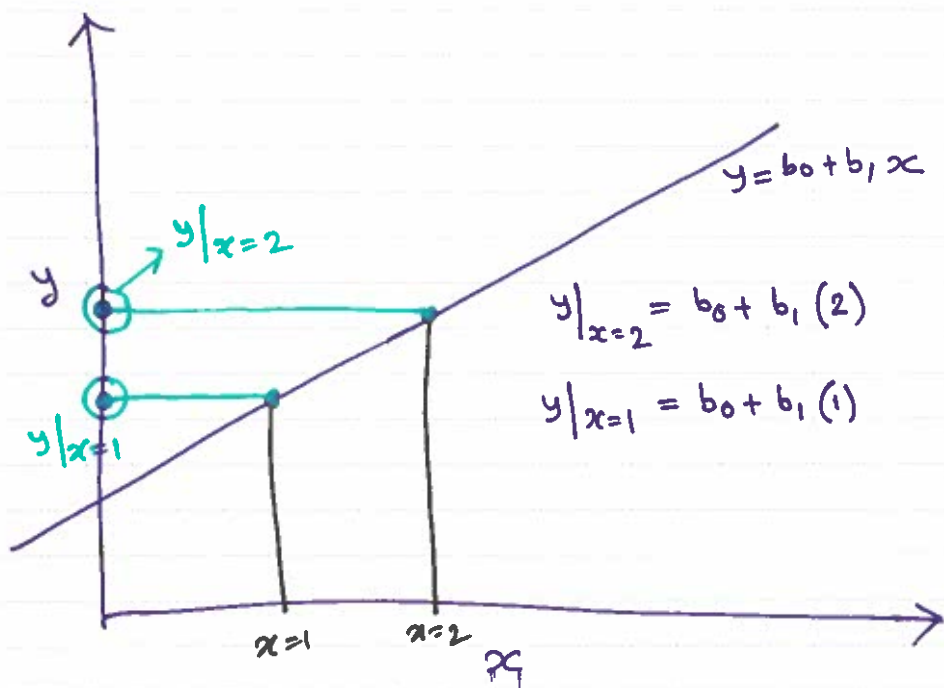
Analytical (closed form) solution

$$\hat{b}_1 = \text{correlation}(x, y) * SD(Y) / SD(X)$$

$$\hat{\beta}_1 = \text{mean}(y) - \hat{\beta}_0 * \text{mean}(x)$$

$$\text{Estimated Price} = 15.2 + 2697.3 * \text{Weight}$$

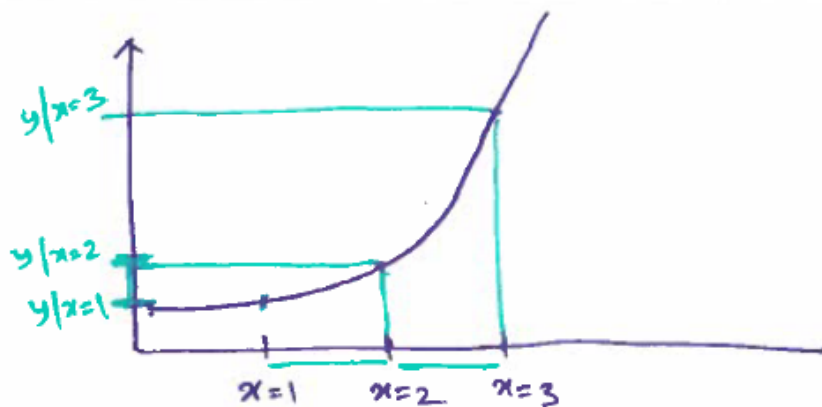
Weight $\uparrow$ 1 unit (carat) $\Rightarrow$ Price $\uparrow$ 2697.3 units(\$)



$$y|x=2 - y|x=1 = b_0 + 2b_1 - (b_0 + b_1) = b_1$$

$$y|x=3 - y|x=2 = b_0 + 3b_1 - (b_0 + 2b_1) = b_1$$

* rate of change in y for a line is
• always fixed



For curves, unlike lines, the rate of change in Y varies at different values of X

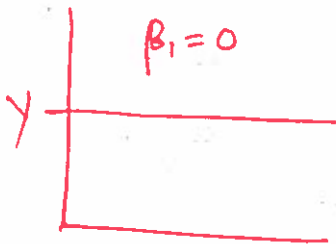
→ coefficient b_0, b_1

LS line from R → will change as my sample changes

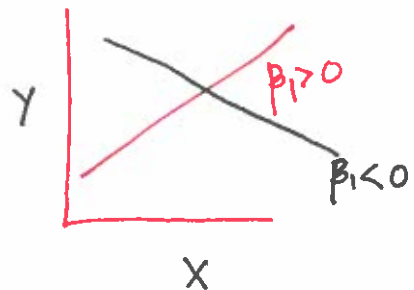
$\left. \begin{matrix} \beta_0 \\ \beta_1 \end{matrix} \right\} \rightarrow \begin{matrix} \text{intercept} \\ \text{slope} \end{matrix}$ of LS line if I had the entire population

$$H_0: \beta_1 = 0 \quad \text{vs} \quad H_A: \beta_1 \neq 0$$

Population: $y = \beta_0 + \beta_1 X$



~~X~~
null



X
alternative

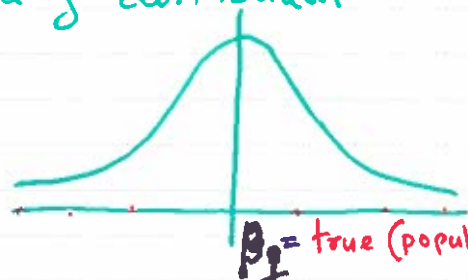
3000 (population)



random sample
 $n = 93$

	b_0	b_1
sample1	.	.
Sample2	.	.
⋮		
Sample10000	.	.

Consider any statistic ($\hat{b}_1$: slope)
as its value changes from sample to sample
the distribution of values over all possible
random samples, then it is called
sampling distribution



Standard Error = Standard deviation of the
Sampling distribution

The sampling distribution of slope is approximately
normal with mean = population line's slope

sd = Std. Error (from R software)

Statistic - mean

Std. Error.

Recall: sample mean

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{SE} = \frac{\bar{X} - \mu_0}{\left(\frac{\sigma}{\sqrt{n}}\right)}$$

Slope

$$\text{statistic} = \frac{\text{coefficient value} - 0}{SE(\text{coefficient})}$$